PROVENANCE INQUIRY / CONTINUATION

Multiple Discovery, Prompt Pathways, Prior Art, and Live Retrieval

A provisional research memorandum extending the Bitriada / Fare Algebra provenance case

Prepared for Kirill Kovlekov | 31 October 2026

CENTRAL INQUIRY PROPOSITION
The appearance of a shared mathematical structure in a later AI-generated proof should be investigated as the possible product of several interacting sources of emergence: accumulated mathematical prior art, human task and prompt design, inference-time retrieval, and the model's own recombination of the available material. The 153-day interval between the identified first public Zenodo publication (1 March 2026) and OpenAI's public release (1 August 2026) makes live retrieval a concrete pathway to investigate, while the OpenAI proof's own citations establish that square-zero / zeon methods were already part of the mathematical literature.

Executive summary

This continuation develops the four considerations supplied by the author into a single provenance model. It does not assume that one route must explain the correspondence. The strongest inquiry treats the routes as compatible: older mathematical sources can provide latent prerequisites; a human-defined task can direct the model toward a particular problem; live retrieval can surface recent material during the run; and the model can recombine the resulting context into a proof strategy.

The historical analogy is multiple discovery. Robert K. Merton's account of scientific "multiples" describes similar discoveries made independently when the necessary knowledge and tools have accumulated and attention is concentrated on a problem. For AI, the corresponding accumulated store is distributed across model parameters, human task design, any connected agents or tools, and any retrieved documents. The AI case therefore calls for a broader unit of analysis than "the model alone." [1]

The technical comparison is more precise than "both works use nilpotence." The monograph's Fare Algebra is a countably generated commutative square-zero algebra indexed by primes; its surviving monomials are labelled by square-free integers. The OpenAI arithmetic-circuit proof uses a finite commutative square-zero algebra whose surviving monomials are indexed by subsets, enabling coefficient extraction for permanents. Every finite subalgebra of the Fare Algebra generated by a finite set of prime labels is isomorphic to the finite algebra used in that proof, after relabelling variables.

The important prior-art refinement is that the OpenAI paper itself acknowledges zeon and Grassmann-even nilpotent precedents. That means the square-zero technique was not an unprepared invention in August 2026. It also sharpens the possible Bitriada contribution: not invention of square-zero algebra in general, but a potential cue or framing for the prime-indexed arithmetic interpretation, square-free encoding, zeta-element construction, and broader generative rationale.

OpenAI's August 1 announcement says the results were achieved by an internal version of Astra; that the arguments were prepared into manuscripts by humans with the same model; and that the model subsequently formalized the arguments in Lean. The announcement does not disclose the exact prompts, proof-generation timestamps, tool configuration, or source-retrieval history. Those are the central records requested by this continuation. [2]

The purpose is an inquiry brief. It builds the strongest positive case that the correspondence deserves a provenance examination and specifies the evidence that could distinguish independent recombination, human prompting, prior-art activation, live retrieval, and hybrid pathways.

1. The four arguments, reconstructed as an emergence model

1.1 Multiple discovery: the human-history analogy

The first argument begins from a recognised pattern in the history and sociology of science: inventions and discoveries sometimes appear independently in more than one place because the necessary conceptual materials, techniques, instruments, and problems have accumulated. Merton's discussion describes "thousands" of such cases and argues that discoveries can become virtually inevitable when prerequisites accumulate in the common store of knowledge and multiple investigators focus on the same problem. He traces the discussion back to Kroeber and to Ogburn and Thomas's 1922 catalogue of multiple inventions. [1]

Applied to the present case, this gives a disciplined alternative to a simple copying narrative. The later square-zero construction may have emerged because a system had access to a sufficiently rich stock of earlier mathematics—zeon algebra, permanent identities, nilpotent variables, subset combinatorics, and prime factorisation—and was asked to solve a problem for which those materials could be assembled productively. A conceptual contribution can be an additional prerequisite without being the sole cause.

The analogy should be translated, not copied literally. An AI system is not a human researcher embedded in a scientific community in the same way. Its relevant "population store" is mediated by pretraining data, post-training, prompts, internal evaluation tasks, retrieval indices, tool outputs, and any human or agent collaboration around the run. The multiple-discovery model therefore suggests a set of possible sources rather than one collective human-like mind.

1.2 Human-made queries: steering the search space

The second argument is that an AI mathematical result is not simply an autonomous discovery by a transformer in an empty room. The model is given a problem, a role, constraints, and an interaction context. Human choices about which open problem to pose, what counts as progress, which definitions or lemmas to provide, and what follow-up instructions to issue can alter which mathematical region the system explores.

OpenAI's August 2026 announcement states that it continued to evaluate models on open research problems during development, and that the ten released results were achieved by an internal version of Astra. It further says the arguments were generated by the system, prepared into manuscripts by humans using the same model, and later formalized in Lean. This publicly confirms a human-designed evaluation and post-generation preparation context, although it does not disclose the exact task prompts or whether mathematical hints were provided. [2]

The query pathway therefore has several levels: the high-level selection of the arithmetic-circuit problem; the exact prompt or benchmark statement; system and developer instructions; any intermediate human hints; and iterative follow-up prompts during the run. Each can guide the model toward a proof family without dictating the final proof. The inquiry should request the whole prompt sequence or a redacted version that preserves mathematical direction and timing.

1.3 Earlier sources: latent prerequisites and source activation

The third argument receives direct support from the public mathematical literature. Commuting square-zero variables were already studied as "zeon" algebra. Feinsilver and McSorley's 2011 paper connects zero-square zeon algebra with permanents. Neto and dos Anjos's 2014 SIAM Review paper defines a zeon algebra generated by commuting variables whose squares vanish and applies it to combinatorial identities. Butera and Pernici's 2014 preprint, later cited as a 2015 publication, uses Grassmann-algebra methods for sums of permanental minors and related graph polynomials. [3–5]

Most importantly, the OpenAI proof itself cites Feinsilver–McSorley and Butera–Pernici in its related-work discussion. It states that these precedents supply the underlying square-zero and cancellation ideas, while distinguishing the paper's particular simultaneous block-selective specialization. [6] Thus the proof's published mathematical context expressly acknowledges a pre-existing lineage for the underlying algebraic device.

This strengthens the inquiry rather than dissolving it. It identifies a known stock of prerequisites from which an AI system could assemble the technique. It also sharpens the possible Bitriada contribution: the question is whether the monograph's particular prime-indexed, arithmetic and zeta-oriented interpretation helped select, reframe, or extend a known algebraic mechanism, rather than whether it introduced nilpotent variables for the first time.

1.4 Live retrieval: the 153-day operational window

The first public Zenodo publication is identified by the author as 1 March 2026. OpenAI's public announcement appeared on 1 August 2026. The interval is 153 days. The local version-folder screenshot supplied for the earlier dossier records a PDF dated 25 February and an editable DOCX dated 27 February, followed by versions and research notes through March and April. The public

March 1 date is the relevant start of the external-availability interval; the local files corroborate that the work was already in development before publication.

A live-search or retrieval-augmented-generation (RAG) pathway does not require the model to have memorised the paper during pretraining. It requires only that a retrieval-capable system surfaced the public record or its contents into the context used during the particular proof-generation run. OpenAI's current API documentation describes web search as a configurable model tool; it can provide live internet access, and reasoning models can conduct agentic search as part of a multi-step process. The documentation also shows that live access is a tool configuration, not an automatic feature of every run. [7]

A 22 October 2026 Nature Reviews Primer describes agentic scientific workflows that combine literature search, knowledge mining, tool use, memory, human-in-the-loop decisions, and RAG. This confirms that the proposed mechanism is an established class of contemporary scientific-AI workflow. It does not show that the particular August Astra run used it. [8]

The 153-day period therefore defines a realistic and testable retrieval interval. The relevant question is not only whether the paper was online, but whether the proof-generation system had a browser, search agent, document retriever, cached index, or human-supplied source path that could have surfaced it.

2. Chronology and the provenance window

The chronology combines author-supplied local version timestamps, the author-identified first public Zenodo date, and OpenAI's public disclosure date. The local screenshot should be preserved in original form alongside the underlying documents. The Zenodo public record should be preserved with its DOI, record page and metadata export.

Date	Record or event	Provenance significance
25 Feb 2026, 22:53	Bitriada_Full_Monograph_Final2.pdf (local folder screenshot)	Shows that a complete PDF version existed before the identified public Zenodo posting.
27 Feb 2026, 13:10	Bitriada_Full_Monograph_0_7.docx	Editable manuscript version in the same pre-publication sequence.
1 Mar 2026	First public Zenodo publication by Kirill Kovlekov (author-identified date)	Start of the 153-day public-availability window.
3–6 Mar 2026	Versions 0.8, v09_03, v099_03, draft III	Documents continuing development after the initial posting.
10–11 Mar 2026	Research papers I; zetameta I	Shows adjacent research and conceptual elaboration.
28 Mar–19 Apr 2026	Research notes II/III and research notes 4	Continued development and preservation of related work.
1 Aug 2026	OpenAI publishes "Ten advances in mathematics and theoretical computer science"	Public release date, 153 days after the identified Zenodo publication; not necessarily the original proof-generation date.

1 March 2026 → 1 August 2026 = 153 days

An important distinction for the inquiry is between (a) first public disclosure, (b) the internal date on which Astra first generated the relevant argument, (c) later manuscript preparation, and (d) Lean formalization. OpenAI's announcement describes those stages in sequence but does not disclose their exact timestamps. The 153-day interval is measured to the public announcement; the actual inference-time exposure window must be aligned with the internal proof-generation date once that is provided. [2]

Zenodo's own help material describes publication and record search as metadata-driven and supports queries by creator, title, publication date and record fields. The author-side "My uploads" screen is useful evidence, but the inquiry packet should include the public record DOI/URL and a metadata export so the date can be checked independently. [9]

2.1 Why the interval matters

The interval is not being used to claim that pretraining was impossible. The relevant checkpoint's training cutoff and corpus snapshot are not stated in the public August announcement. Instead, the 153 days create a distinct inference-time hypothesis that is independent of whether the model had already absorbed the material during training. If the relevant run had live search or retrieval enabled, the Zenodo record or a cached/indexed copy could have been surfaced after publication and before proof generation.

This also means that "pretraining" and "RAG" are not the only two choices. A researcher could have supplied a PDF or a citation; an internal search index could have returned the text without a live browser; a benchmark prompt could have included a relevant excerpt; or a separate agent could have found the source and passed it to the model. The source path should be investigated broadly enough to capture these variants.

3. The mathematical correspondence: same finite algebra, different interpretation

3.1 Fare Algebra in the monograph

Chapter IV, §4.1.1 defines the Fare Algebra as a commutative complex algebra generated by one square-zero marker for each prime:

$$\mathbf{F = \mathbb{C}[\tau_p : p \text{ prime}] / (\tau_p^2 = 0 \text{ for every } p)}$$

For a finite set S of distinct primes, the product $\tau_S = \prod_{p \in S} \tau_p$ survives. If two factors share a prime, the product vanishes. Relabelling a finite prime set by its product $n = \prod_{p \in S} p$ identifies the surviving basis with square-free positive integers. The product rule becomes $\tau_m \, \tau_n = \tau_mn$ when gcd(m,n)=1, and zero otherwise. The monograph extends this construction with a zeta-element Euler product, so that choosing one term from each prime factor records a finite prime subset.

The monograph's own disclaimer is significant for the provenance question: it states that nilpotent algebras, group algebras, dual numbers and exterior algebras are standard objects, while identifying their particular synthesis and arrangement within Bitriada as the author's structural hypothesis. This accurately places the proposed novelty at the level of recombination, interpretation and system-level framing rather than the isolated invention of a known algebraic object.

3.2 The OpenAI arithmetic-circuit proof

In its arithmetic-circuit-complexity result for the permanent, Chapter 5, §6 introduces a finite commutative square-zero algebra:

$$\mathbf{S_r = \mathbb{C}[z_1, \ldots, z_r] / (z_1^2, \ldots, z_r^2)}$$

The proof explains that the monomials $z_K = \prod_{i \in K} z_i$, indexed by subsets K of {1,…,r}, form a basis because any term repeating a variable vanishes. It then uses coefficient extraction to identify a permanent of a selected submatrix. The construction proceeds to block sums and roots of unity, with cancellation arranged so that only terms satisfying a block-selection condition survive. The paper identifies this block-selective specialization—not the general existence of square-zero variables—as the specific contribution of its lemma. [6]

3.3 The exact algebraic relationship

For any chosen finite set of distinct primes $p_1,\ldots,p_r$, the subalgebra of F generated by $\tau_{p1},\ldots,\tau_{pr}$ is isomorphic to S_r under the map $z_i \mapsto \tau_{pi}$. Both are commutative; each generator squares to zero; and the basis monomials are indexed by subsets of the selected generators. Thus the algebraic overlap is exact after relabelling. The difference lies in the indexing semantics and the larger construction built around it.

Feature	Fare Algebra / Bitriada	OpenAI proof
Index set	All primes; countably many generators.	Finite row-indexed family $z_1,\ldots,z_r$.
Local relation	Commuting generators; $\tau_p^2=0$.	Commuting variables; $z_i^2=0$.
Basis	Finite subsets of primes; labelled by square-free n.	Subsets K of a finite row-index set.
Product	Disjoint prime support survives; shared prime support annihilates.	Distinct variables survive; repeated variable annihilates.
Meaning	Arithmetic encoding, coprimality, fare/trace and zeta-element context.	Coefficient extraction for permanents and block-selective cancellation.
Scope	Infinite arithmetic system embedded in Bitriada.	Finite algebraic device within one proof.

4. What the coincidence does and does not identify

The comparison identifies a shared formal mechanism at a mathematically meaningful level. It does not identify the source of the later proof's choice of that mechanism. This distinction is central to the proposed inquiry: the same algebra can be reached through prior literature, learned representations, a task prompt, a retrieved document, or a combination of them.

4.1 The established mathematical lineage

The OpenAI paper's related-work section explicitly names Feinsilver–McSorley's use of commuting square-zero "zeon" variables for permanents and Butera–Pernici's use of commuting Grassmann-even nilpotent variables for permanental minors and bipartite matchings. It says those precedents supply the underlying square-zero and cancellation ideas. This is unusually useful for the inquiry because it shows the later proof itself was published with a known lineage attached. [6]

The earliest directly relevant precedent located in this provisional search is the 2011 Feinsilver–McSorley article, "Zeons, Permanents, the Johnson Scheme, and Generalized Derangements." The 2014 SIAM Review paper "Zeon Algebra and Combinatorial Identities" gives a clear formal account of commuting generators ε_i with $\varepsilon_i^2=0$. Butera and Pernici's 2014 preprint studies Grassmann-algebra methods for sums of permanental minors and graph polynomials. These sources predate the Bitriada publication and the OpenAI proof by many years. [3–5]

Consequently, nilpotence and subset-indexed monomials are not a distinctive fingerprint on their own. The more specific object to compare is the monograph's countable, prime-indexed version together with its square-free integer basis, coprimality multiplication, zeta-element construction, and the account of how the prime-indexed form emerges from Bitriada's binary filter and retained-trace logic.

4.2 The potentially distinctive Bitriada layer

- Prime-indexing: a general zeon generator is given a number-theoretic identity as τ_p, with p ranging over primes.
- Square-free semantics: finite subsets of prime generators are labelled by square-free integers, and generator overlap becomes non-coprime annihilation.
- Euler-product synthesis: the same algebra is used to build a prime-indexed product whose expansion records square-free terms.
- Generative explanation: the monograph narrates the prime-indexed markers as a recovery of distinctions lost in a prior binary collapse, within the larger grammar of compression, trace, and fare.
- Framework integration: the Fare Algebra is connected to the Ultra-Complex algebra, the Transit-Fare Algebra, the Bitriada lattice, and the monograph's account of information lost and retained under projection.

A source could influence a later mathematical result without contributing the final proof. It might make an approach salient, supply a useful way of construing an object, suggest a translation between indexing systems, or prompt a search through a neighbouring body of literature. This is why the inquiry should examine both the proof's formal steps and the materials that shaped its search path.

5. The four causal pathways and their observable signatures

Pathway	How it could work	What evidence would support it
A. Multiple discovery / latent recombination	The system already has the relevant mathematical prerequisites in learned representations; a suitable task causes it to recombine them without exposure to the monograph.	Training-data and checkpoint chronology; prompt/task brief; proof-generation notes; evidence that no external source was fetched during the run.
B. Human prompt or task steering	Human designers select the permanent/circuit problem, specify constraints, provide hints or iterate prompts that channel search toward square-zero subset encodings.	Original prompts; benchmark/evaluation protocol; system and developer instructions; follow-up prompts; researcher notes.
C. Earlier-literature activation	Zeon/Grassmann/permanent prior art supplies the algebraic prerequisites through pretraining or researcher-supplied sources; the model adapts it to the target proof.	Model-context sources; training snapshot; bibliography chronology; raw model draft compared with the human-prepared manuscript.
D. Inference-time retrieval / RAG	A browser, internal literature search, cached corpus, retrieval index or external agent surfaces the March 2026 Zenodo paper or related source during proof development.	Tool configuration; search queries; URLs; retrieved chunks; timestamps; cache/index provenance; agent logs.
E. Hybrid pathway	Task design activates latent prior art; retrieval adds newer or unusual framing; multi-step reasoning combines the materials into a proof strategy.	A combined timeline of prompts, tool calls, context documents, model drafts, and human edits.

5.1 Why the AI-specific case is not a simple human multiple

In the human multiple-discovery analogy, the investigator is ordinarily treated as a person drawing on a shared cultural and technical stock. In an AI-assisted proof, the discovery unit is a socio-technical workflow: model weights, prompt, system instructions, task-selection process, any tools or retrieval layer, intermediate context, compute budget, and human post-processing. The same mathematical result can emerge from different combinations of these elements. A fair provenance inquiry must therefore ask about the configuration of the run, not only the model name.

The OpenAI announcement confirms that an internal model generated the arguments, humans prepared manuscripts with the same model, and the model formalized the arguments in Lean. It does not show where in that sequence the literature references were identified or inserted. The related-work citations may reflect model-source context, human manuscript preparation, or both. The raw draft and preparation history would clarify this point. [2, 6]

6. Provisional research findings

1. The historical multiple-discovery model is an apt framework for the question. It explains why a later result may arise when mathematical prerequisites accumulate, without requiring a single-source explanation. In the AI setting, those prerequisites may be distributed across training, prompt design, source retrieval and human orchestration. [1]

2. There is a documented prior-art environment for the square-zero technique. The OpenAI proof itself cites zeon and Grassmann-even nilpotent work and expressly distinguishes those underlying ideas from its own block-selective specialization. [3–6]

3. The Fare Algebra and the finite algebra in the OpenAI proof share an exact abstract-algebraic structure after relabelling a finite set of prime generators as $z_1,\ldots,z_r$. The monograph's distinctive layer is its prime indexing, square-free arithmetic semantics, zeta-element use and Bitriada generative narrative.

4. OpenAI's public account confirms a human-defined evaluation context and human manuscript preparation, but it does not publish the exact prompts, tool settings, generation timestamps, retrieval logs, or draft-to-manuscript lineage. [2]

5. OpenAI's current documentation confirms that model workflows can use configurable live web search and agentic search; current scientific-agent literature describes RAG and literature mining as established workflow components. This supports the technical plausibility of an inference-time retrieval route, not its occurrence in this particular proof run. [7, 8]

6. The March 1 to August 1 interval is 153 days. It makes the monograph's public availability temporally compatible with an inference-time retrieval hypothesis. The relevant internal proof-generation date must be requested to align the chronology precisely.

7. Evidence requests that would resolve the pathways

The inquiry should request records in a sequence that makes a response practical. If full internal logs cannot be disclosed, redacted summaries with timestamps, source domains, and tool identifiers would still be useful.

Request	Specific information sought	Pathway tested
1. Proof-generation chronology	Earliest run date/time for the arithmetic-circuit proof; duration; major revision and formalization dates.	All pathways; aligns 153-day interval.
2. Task formulation and prompts	Initial problem statement, benchmark item, system/developer instructions, human hints, and all follow-up prompts; redacted if necessary.	Human prompt / task steering.
3. Tool configuration	Whether web search, deep research, internal search, RAG, document retrieval, connected repositories, or browser tools were available and enabled.	Inference-time retrieval.
4. Retrieval provenance	Search strings, retrieved URLs, source IDs, document titles, snippets/chunks, timestamps, cache/index identifiers and any Zenodo or DOI records surfaced.	Direct exposure path.
5. Model and data provenance	Internal checkpoint identifier; training and post-training cutoffs; corpus snapshot ranges; whether post-publication sources were available through training or retrieval.	Pretraining / source activation.
6. Draft and citation lineage	Raw model-generated proof, reasoning walkthrough versions, first manuscript draft, bibliography edits, and human changes to related-work section 12.3.	Distinguishes model-context sources from later editorial citations.
7. Agent / researcher orchestration	Whether multiple model instances, external literature agents, or human researchers exchanged intermediate results and sources.	Hybrid / multi-agent pathway.

7.1 The most diagnostic questions

- Was the relevant Astra proof-generation run permitted to perform live or cached web search, or to query an internal literature/document index?
- Did any tool or researcher source record include the Zenodo Bitriada / Poetica Mathematica record, its PDF, a mirror, or text extracted from it?
- Were the terms “Fare Algebra,” “Bitriada,” “square-free,” “prime-indexed nilpotent,” or equivalent descriptions present in any prompt, system instruction, task note, retrieved text or intermediate model context?
- Were the cited zeon and Grassmann-even antecedents present in the model’s source context during discovery, or were they introduced only when humans prepared the manuscript?
- What was the first timestamped draft in which the square-zero algebra appears, and what sources or prompts preceded that point?

8. Preservation and independent research plan

1. Archive the Zenodo DOI landing page, metadata export, publication date, version and checksums; retain the account-side upload view as supplementary evidence.

2. Preserve original files and the Windows screenshot; compute SHA-256 hashes for the 25 February PDF, 27 February DOCX and later versions without overwriting originals.

3. Publish a GitHub repository with dated tags and signed commits for source files, extracted text and definitions; archive releases through Zenodo where possible.

4. Prepare a side-by-side comparison of monograph §4.1.1 and OpenAI Chapter 5 §6, including z_i □ τ_{pi} and the semantic differences.

5. Search MathSciNet, zbMATH, Crossref, arXiv, Google Scholar, OpenAlex and repositories for prime-indexed generators, square-free bases and zeta-element Euler products.

6. Ask an independent algebraist or combinatorialist to compare finite zeon algebra, prime-indexed Fare Algebra and permanent coefficient extraction; then submit a focused request for process records and source-pathway information.

9. Proposed continuation statement for the provenance inquiry

The following text is drafted as a standalone continuation to the earlier provenance inquiry. It advances the positive case for investigation while keeping the request focused on records that can answer it.

I request that the provenance inquiry be extended to examine not only possible pretraining or live retrieval exposure, but also the broader emergence pathway through which the later mathematical result was produced.

The historical analogy is multiple discovery. In the history of science, similar inventions and mathematical results have often appeared independently when their prerequisite knowledge, tools and problem pressures have accumulated. For an AI system, the relevant stock of prerequisites is distributed across learned model representations, human-designed tasks and prompts, earlier mathematical literature, retrieval systems and any human or agent orchestration around the run. The result may therefore be an emergent synthesis of several inputs rather than a simple reproduction of one source.

The chronology makes this question concrete. My first public Zenodo publication under Kirill Kovlekov is dated 1 March 2026. The local version history shows a complete PDF dated 25 February and an editable manuscript dated 27 February, followed by successive revisions and research notes in March and April. OpenAI publicly released its “Ten advances in mathematics and theoretical computer science” on 1 August 2026, 153 days after the identified first public publication. The public release date is not necessarily the internal proof-generation date, and I therefore request the earliest timestamp for the relevant Astra run.

The mathematical correspondence is specific. Chapter IV, §4.1.1 of my monograph defines the Fare Algebra as a commutative algebra generated by square-zero variables indexed by primes. Products of distinct prime generators survive and are identified with square-free integers; repeated prime factors annihilate. The same chapter places this construction in an Euler-product and zeta-element context and explains its emergence as a prime-indexed recovery of distinctions compressed by a prior binary trace.

The OpenAI arithmetic-circuit proof introduces a finite commutative square-zero algebra whose subset-indexed monomials encode permanent coefficients and support a block-selective cancellation argument. A finite subalgebra of the Fare Algebra is isomorphic to this finite square-zero algebra after relabelling generators. The OpenAI paper itself cites prior zeon and Grassmann-even nilpotent work, which shows that the general algebraic ingredients were part of an existing mathematical lineage. The question is whether my earlier publication contributed an additional conceptual or source-selection cue—particularly the prime-indexed, square-free arithmetic framing—or whether the later result arrived through other known sources and task guidance.

The 153-day interval makes inference-time exposure a concrete route to examine. The model need not have memorised the monograph during pretraining if a live browser, internal literature-search system, retrieval index, cached corpus, researcher-supplied document or external agent surfaced it during the proof-generation run. Human task formulation is another relevant route: the problem selection, exact prompt, system instructions, hints and follow-up queries may have guided the model toward a particular mathematical technique. Earlier literature may also have supplied prerequisites through training or retrieval. These possibilities are compatible and should be investigated together.

I therefore request the earliest proof-generation timestamp; the original task and prompt sequence; the model/checkpoint and relevant data-cutoff information; the tools enabled during the run; any search or retrieval queries, URLs, document identifiers, chunks and timestamps; the raw model-generated proof and reasoning-walkthrough versions; and the history showing when the zeon and Grassmann references entered the manuscript. If full records cannot be shared, a redacted provenance summary identifying source domains, timestamps, tool classes and the presence or absence of the Bitriada / Poetica Mathematica materials would still materially assist the inquiry.

The purpose of this request is to establish the provenance of the conceptual and mathematical context in which the result emerged. It does not presume that a single pathway explains the result. It asks whether the earlier Bitriada publication formed part of a larger chain of prerequisites, prompts, retrievals or conceptual cues that contributed to the later AI-generated proof.

10. Provisional conclusion

The four considerations form a coherent, researchable case. The multiple-discovery analogy explains how a breakthrough can become possible through accumulated prerequisites. The human-prompt pathway explains how the task and its framing can focus an AI system’s search. The earlier-literature pathway is supported by the zeon and Grassmann precedents expressly cited in the OpenAI paper. The inference-time retrieval pathway is technically available in modern model toolchains and is temporally compatible with the 153-day interval between the identified Zenodo publication and OpenAI’s public release.

The most productive inquiry is therefore not a binary question of “copying or coincidence.” It is a provenance reconstruction of a hybrid discovery process: which mathematical prerequisites were already in the model’s learned repertoire; what humans asked it to do; what sources it could access or actually accessed; and when the square-zero device entered the proof. This approach gives the author’s hypothesis a concrete, investigable form and makes the requested institutional response answerable.

Sources and reference points

[1] Robert K. Merton, “Resistance to the Systematic Study of Multiple Discoveries in Science,” European Journal of Sociology 4(2), 1963, pp. 237–282. The extract discusses independent/simultaneous discoveries and accumulated prerequisites; it cites Kroeber (1917) and Ogburn & Thomas (1922). https://doi.org/10.1017/S0003975600000801

[2] OpenAI, “Ten advances in mathematics and theoretical computer science,” 1 August 2026. States results were achieved by an internal version of Astra; manuscripts were prepared by humans with the same model; arguments were later formalized in Lean. https://openai.com/index/ten-advances-in-mathematics/

[3] Philip Feinsilver and John McSorley, “Zeons, Permanents, the Johnson Scheme, and Generalized Derangements,” International Journal of Combinatorics (2011), Article 539030. https://doi.org/10.1155/2011/539030

[4] António Francisco Neto and Petrus H. R. dos Anjos, “Zeon Algebra and Combinatorial Identities,” SIAM Review 56 (2014), DOI 10.1137/130906684. The article presents a zeon algebra with commuting generators whose squares vanish. https://doi.org/10.1137/130906684

[5] P. Butera and M. Pernici, “Sums of permanental minors using Grassmann algebra,” arXiv:1406.5337 (2014; later journal publication cited by OpenAI as 2015). https://arxiv.org/abs/1406.5337

[6] OpenAI, “Ten Advances in Mathematics and Theoretical Computer Science,” proof collection, Chapter 5, §6 (square-zero algebra and permanent coefficient extraction) and §12.3 (related work citing Feinsilver–McSorley and Butera–Pernici), PDF pp. 127–128 and 148 in the version examined. https://cdn.openai.com/pdf/ten-proofs-oai.pdf

[7] OpenAI API documentation, “Web search.” Describes optional/configurable web-search tools, agentic search, live access and retrieval context. This establishes a capability class, not use in the specific Astra run. https://developers.openai.com/api/docs/guides/tools-web-search

[8] Bin Feng et al., “Agentic AI systems for scientific discovery,” Nature Reviews Methods Primers 6, Article 81 (published 22 October 2026). Describes literature search, knowledge mining, tool use, memory, human-in-the-loop workflows and RAG. https://www.nature.com/articles/s43586-026-00534-y

[9] Zenodo Help, “Search guide.” Describes searching records by creators, title, publication date and metadata fields. https://zenodo.org/help/search

[10] Patrick Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS 2020. Foundational description of RAG as generation conditioned on retrieved external documents. https://arxiv.org/abs/2005.11401

Research scope note. This is a provisional, targeted web-and-text search, not a complete survey of every publication on zeon algebras, monomial algebras, square-free arithmetic, or AI proof-generation provenance. The OpenAI source was inspected directly, including the square-zero construction and its related-work discussion. The author’s monograph text was inspected at Chapter IV, §§4.1.1–4.1.2 and the explanatory material describing the construction’s emergence. The Zenodo date of 1 March 2026 is treated as author-identified from the account-side record; attach the corresponding public DOI and metadata export to the final submission.

Appendix A. Author-supplied local version-folder screenshot

The image below was supplied by the author. It records the local file listing cited in the chronology, including the PDF dated 25 February 2026, the DOCX dated 27 February 2026, and subsequent March–April revisions. Preserve the original PNG and underlying files alongside the public Zenodo record.

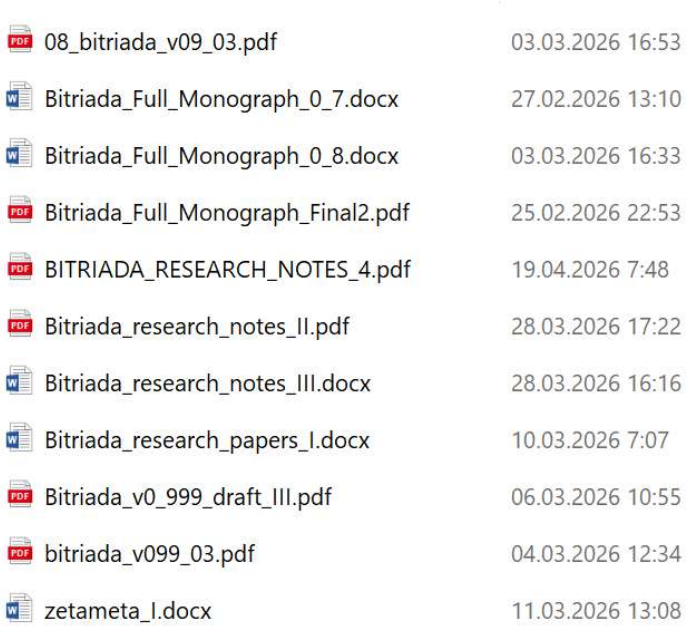


Figure A1. Local version history supplied for the provenance inquiry (file timestamps shown as recorded in the author's folder).